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Human Kinetics
ORIGINAL INVESTIGATION

Cluster Sets: Permitting Greater Mechanical Stress Without Decreasing Relative Velocity

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Purpose: To determine the effects of intraset rest frequency and training load on muscle time under tension, external work, and external mechanical power output during back-squat protocols with similar changes in velocity. ***Methods:*** Twelve strength-trained men (26.0 ± 4.2 y, 83.1 ± 8.8 kg, 1.75 ± 0.06 m, 1.88:0.19 one-repetition-maximum [1RM] body mass) performed 3 sets of 12 back squats using 3 different set structures: traditional sets with 60% 1RM (TS), cluster sets of 4 with 75% 1RM (CS4), and cluster sets of 2 with 80% 1RM (CS2). Repeated-measures ANOVAs were used to determine differences in peak force (PF), mean force (MF), peak velocity (PV), mean velocity (MV), peak power (PP), mean power (MP), total work (TW), total time under tension (TUT), percentage mean velocity loss (%MVL), and percentage peak velocity loss (%PVL) between protocols. ***Results:*** Compared with TS and CS4, CS2 resulted in greater MF, TW, and TUT in addition to less MV, PV, and MP. Similarly, CS4 resulted in greater MF, TW, and TUT in addition to less MV, PV, and MP than TS did. There were no differences between protocols for %MVL, %PVL, PF, or PP. ***Conclusions:*** These data show that the intraset rest provided in CS4 and CS2 allowed for greater external loads than with TS, increasing TW and TUT while resulting in similar PP and %VL. Therefore, cluster-set structures may function as an alternative method to traditional strength- or hypertrophy-oriented training by increasing training load without increasing %VL or decreasing PP.

Keywords: intraset rest, fatigue, velocity, hypertrophy, strength, power

Traditional periodization paradigms normally contain blocks of training that target specific adaptations. From a coaching standpoint, some training periods may include high volumes of fatiguing resistance training to increase skeletal-muscle hypertrophy, with little concern for the deleterious effects of fatiguing exercise on other components of fitness, such as power output. On the other hand, the application of parallel models of periodization allow for, and may even require, a multifaceted approach to training in which hypertrophy, strength, and power output are all considered throughout a microcycle. Therefore, there is a need for research examining different resistance-training schemes that aim to simultaneously develop a combination of hypertrophy, strength, and external mechanical (or system) power output.

As a result, researchers have investigated strategies that allow for the maintenance of acute system power output during exercises that are normally performed to elicit strength or hypertrophic adaptations. For example, it has been shown that power output decreases when performing high-volume back squats using traditional sets (ie, performing repetitions in a set without intraset rest), but such decreases are ameliorated when using cluster sets (ie, including intraset rest intervals).[1–3] Most of the cluster-set literature includes protocols in which the external load and the number of repetitions are kept constant between the experimental (cluster-set) and control (traditional-set) protocols,[4–6] making the external load and training volume controlled independent variables and fatigue-induced changes in velocity a dependent variable of interest. However, no research has intentionally prescribed different external loads for the same number of repetitions during traditional and cluster-set structures to make decreases in concentric velocity a controlled independent variable. In doing so, investigators would be able to determine how cluster-set manipulation can allow for increases in training load and the resultant acute power-output responses.

If the external resistance is increased, movement velocity will decrease and the ensuing power output may rely more heavily on force production if power output is to be maintained or increased. Since this hypothesis is currently untested in the cluster-set literature, the purpose of this study was to determine the effects of intraset rest configuration and external load on total work (TW), time under tension (TUT), and power output compared with traditional sets using lighter loads with similar changes in velocity. Based on previous research[1,3,7,8] and the force–velocity relationship, we hypothesized that cluster sets with greater loads would result in slower movement velocities and greater TW and TUT but similar system power outputs than traditional sets with lesser load and similar changes in velocity.

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Methods

Subjects

Twelve strength-trained men (26.0 ± 4.2 y, 83.1 ± 8.8 kg, 1.75 ± 0.06 m) participated in this study and could perform a back squat with the top of the thighs below parallel with a minimum of 150% body mass (mean handheld-goniometry knee-flexion angle of 120.8° ±

12.4°). The average free-weight back-squat 1RM was 153.4 ± 18.4 kg, resulting in a ratio or 1-repetition maximum [1RM] to body mass of 1.88 ± 0.19. Subjects were screened using medical history questionnaires and were excluded if they reported any recent musculoskeletal injuries. All procedures were carried out in accordance with the Declaration of Helsinki, and the university human research ethics committee approved the research study. All subjects gave written informed consent before participation.

Study Design

Subjects reported to the laboratory for a 1RM session and 3 randomized experimental sessions, each of which included 1 of the following protocols: 3 traditional sets (TS) of 12 using 60% 1RM with 120 seconds of seated interset rest; cluster sets of 4 (CS4), inclusive of 3 sets of 12 using 75% 1RM with 120 seconds of seated interset rest and 30 seconds of unloaded standing intraset rest after the fourth and eighth repetitions of each set; or cluster sets of 2 (CS2), inclusive of 3 sets of 12 using 80% 1RM with 120 seconds of seated interset rest and 30 seconds of unloaded, standing intraset rest after the second, fourth, sixth, eighth, and tenth repetitions of each set (Figure 1).

The use of novel protocols, designed to result in similar changes in velocity, required similar relative loading intensities between protocols. Therefore, external loads were assigned at the same relative intensity for each protocol to avoid repetition failure and to provide loading consistency between protocols. Briefly, a repetition-maximum load of the number of repetitions performed in sequence was multiplied by a factor of 0.84, resulting in a "moderate" load being used for all conditions.[9]

Methodology

Repetition-Maximum Testing: Session 1. After anthropometric measurements, subjects cycled on an ergometer for 5 minutes at 60 rpm with 100 W of resistance. Next, 10 body-weight squats were performed, followed by 8, 5, and 3 repetitions at 25%, 50%, and 60% estimated 1RM, respectively. Back-squat 1RM was assessed starting at 85% estimated 1RM and was progressively increased until the 1RM was achieved.[10] Subjects' heel and toe locations were recorded on the force plate using a horizontal-vertical grid intersecting every 1 cm, and foot placement was maintained during all testing sessions. Sessions were separated by 48 to 96 hours, and subjects were instructed to refrain from any type of fatiguing lower-body activity for the duration of the study.

Experimental Testing: Sessions 2, 3, and 4. During these randomized sessions, the warm-up was the same as session 1 but included warm-up squats using the actual percentages of 1RM. To begin each protocol, an investigator provided verbal 10-second countdown and the subject unracked the bar when the countdown reached "zero." After stepping backward onto the force plate and completing the prescribed number of consecutive repetitions (2 repetitions in CS2 or 4 in CS4), subjects reracked the bar in the squat rack and the intraset rest timer started. Subjects stepped backward and remained standing unloaded on the force plate during the 30-second intraset rest period. After 20 seconds, the next 10-second countdown started, and when it reached 5, subjects stepped forward and positioned themselves under the bar, ready to unrack the bar when the countdown reached zero. Repetitions during TS were performed in succession, with no intraset rest. After completing the twelfth repetition in all protocols, subjects racked the bar and the 120-second interset rest timer started. Subjects then walked to a chair 3 steps in front of the force plate and remained seated until the next 10-second countdown began to start the second set. This process was continued until the protocol was complete.

To maximize concentric velocity during the back squat, subjects were instructed to "explode out of the bottom" and perform the concentric phase of each repetition as quickly as possible, back to a standing position.[7] Subjects were instructed to "squat all the way down" while constantly lowering the barbell under control. Squat depth was monitored using live visual displacement curves to ensure that all repetitions were performed to approximately the same depth (no significant differences in mean squat depth within subjects, between conditions: CS2 = 60.12 ± 5.83, CS4 = 60.34 ± 6.30, and TS = 62.37 ± 6.01 cm; $F = 3.20$, $P = .061$). The feet were required to maintain contact with the force plate at all times[7] (eg, no jumping or lifting of the heels) and a slight pause was required after every repetition to encourage full hip and knee extension.

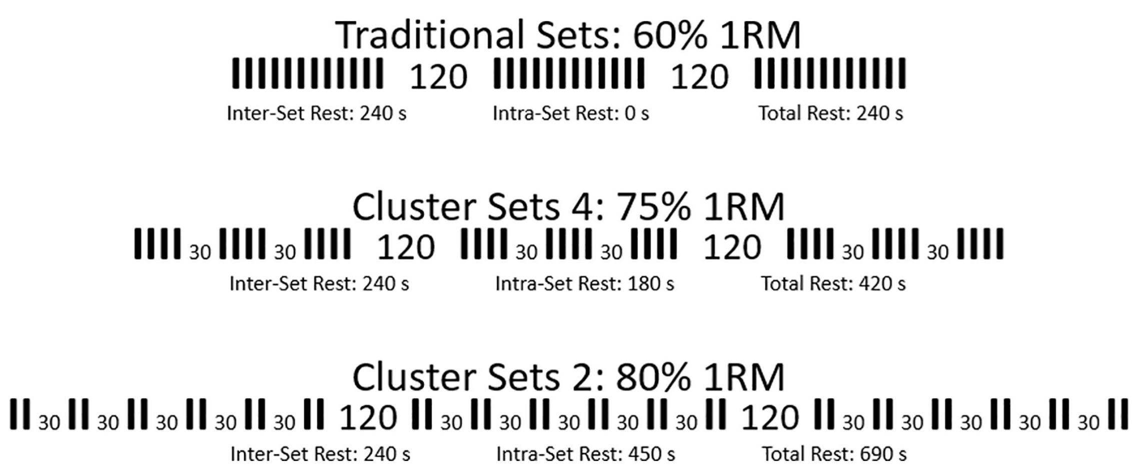

Figure 1 — Protocol designs for 3 sets of 12 repetitions using traditional sets, cluster sets of 4, and cluster sets of 2 inclusive of 120 seconds of seated interset rest and 30 seconds of standing but unloaded intraset rest. Abbreviation: 1RM, 1-repetition maximum.

Data Acquisition. A customized LabVIEW program (National Instruments, Version 14.0) was used to collect and manually analyze data received from the force plate (AMTI BP12001200; Watertown, MA) and linear position transducers (Celesco PT5A-250; Chatsworth, CA) via a BNC-2090 interface box with an analog-to-digital card (NI-6014; National Instruments, Austin TX, USA). All signals were sampled at 1000 Hz and filtered using a fourth-order low-pass Butterworth filter with a cutoff frequency of 50 Hz. The retraction tension of the 4 linear position transducers was 23.0 N, which was accounted for in all calculations.

All kinematic and kinetic data were collected using methodology similar to previous research, and all variables were collected during the concentric phase of each lift unless noted otherwise.[3] Briefly, all squats were performed on a force plate to obtain mean force (MF) and peak force (PF), and 2 linear position transducers were attached to each side of the barbell (4 in total) originating from the top of the squat rack to obtain mean velocity (MV), peak velocity (PV), and vertical displacement of the barbell. External mechanical mean power (MP) and peak power (PP) of the system were calculated by direct measurement of ground-reaction force and bar velocity. The amount of TW was calculated by integrating the area under the force-displacement curve during the eccentric and concentric portions of each repetition.[11] Muscle TUT was computed by summing the time during the eccentric and concentric phases of each repetition.[11] Specifically, the eccentric phase was defined from the moment that descent occurred until maximal (negative) displacement, while the concentric phase was defined as the point of maximal displacement to zero displacement (standing). Since the protocols in the current study included different loads resulting in different absolute velocities, relative velocities were used[12] to determine the percentage change in mean velocity loss (%MVL) and peak velocity loss (%PVL), which were calculated according to Sánchez-Medina and Gonzalez-Badillo[13] using the equation in Figure 2a, adapted for cluster-set protocols. Sánchez-Medina and Gonzalez-Badillo[13] calculated the percentage loss of mean propulsive velocity from the fastest (usually the first) to the slowest (last) repetition of each set and then averaged over the 3 sets. However, the final 4 repetitions of each protocol were averaged together to represent what was occurring near the end of each protocol since the cluster-set structures did not follow a linear decrease in velocity as seen in TS (Figure 2b) and in the visual example provided by Sánchez-Medina and Gonzalez-Badillo.[13] The average %MVL ranged from 14% to 16%, and %PVL ranged from 10% to 13%, indicating that %VL was controlled between protocols.

Fig. 2a

$$MVFT = \left(\frac{AvgMeanVelocity_{Reps\ 33-36} - MeanVelocity_{Rep\ 1}}{MeanVelocity_{Rep\ 1}} \right) * 100$$

Fig. 2b

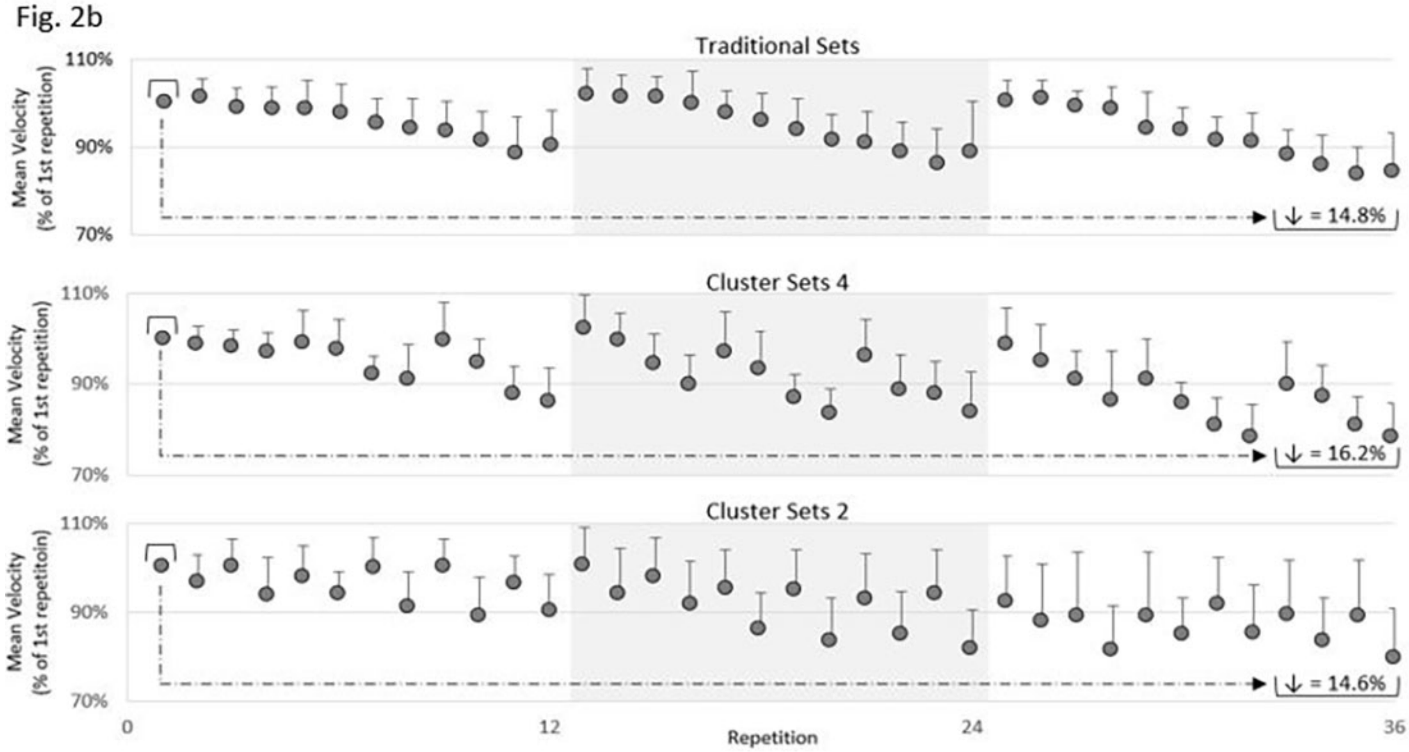

Figure 2 — (a) Equation used to determine percentage mean and peak velocity loss. (b) Example of percentage mean velocity loss calculated by dividing the average of the final 4 repetitions by the first repetition. Percentage decrease in mean velocity (↓) shown in brackets for each protocol. No significant differences between groups.

Statistical Analyses. Means and SDs were calculated for all dependent variables and protocol time. For each variable, a repeated-measures ANOVA was used and a Holm sequential Bonferroni follow-up test was used to control for type I error. Significance was set at $P \leq .05$ for all tests. In addition, effect sizes ± 90% confidence intervals were calculated using Cohen *d* and can be interpreted as small (0.2–0.49), moderate (0.5–0.79), and large (≥0.8). All statistical analyses were performed using SPSS version 22.0 (IBM, Armonk, NY).

Results

As intended by design, the total time taken to complete each protocol was different ($F = 2613$, $P < .001$). Specifically, CS2 (15:51 ± 0:49 min) was greater than CS4 (10:10 ± 0:38 min; $P < .001$, $d = 7.46$) and TS (6:01 ± 0:19 min; $P < .001$, $d = 15.40$), while CS4 was also greater than TS ($P < .001$, $d = 7.62$). There were also significant differences between protocols for MF, MV, PV, MP, TW, TUT, eccentric TUT, and concentric TUT (Table 1). There were no significant differences between protocols for PF or PP (Table 1). For PF, there was a moderate effect for CS2 and TS, a small effect for CS4 and TS, and no effect for CS2 and CS4 (Figure 3). For PP, there were no effects for CS2 and TS, CS4 and TS, or CS2 and CS4, (Figure 3). Effect sizes for MF, MV, PV, MP, TW, and TUT are also shown in Figure 3.

Discussion

The major findings of the current study were that (1) cluster-set structures allowed greater external loads to be lifted than traditional sets while resulting in similar %MVL and %PVL, (2) using greater loads resulted in slower MV but resulted in greater TW and TUT

Table 1 Variables During Each Protocol When Averaged Across All 36 Repetitions, Mean ± SD

	CS2	CS4	TS	ANOVA result
Mean force (N)	1968.25 ± 212.63***†††	1884.91 ± 210.79***	1656.40 ± 184.51	$F = 725.07$, $P < .001$
Peak force (N)	2604.76 ± 250.00	2588.79 ± 301.45	2463.92 ± 257.32	$F = 4.51$, $P = .023$^
Mean velocity (m/s)	0.49 ± 0.06***††	0.54 ± 0.06***	0.69 ± 0.07	$F = 119.90$, $P < .001$
Peak velocity (m/s)	0.99 ± 0.15***†	1.04 ± 0.17***	1.16 ± 0.16	$F = 29.86$, $P < .001$
Mean power (W)	944.06 ± 121.19***†	1006.09 ± 108.07***	1114.11 ± 109.18	$F = 31.23$, $P < .001$
Peak power (W)	2119.79 ± 361.16	2169.90 ± 412.82	2120.50 ± 353.67	$F = 0.63$, $P = .543$
Total work (kJ)	87.89 ± 8.59***†	84.45 ± 8.03***	76.73 ± 7.44	$F = 30.93$, $P < .001$
TUT (s)	87.69 ±10.35***††	80.58 ± 10.00**	69.55 ± 6.82	$F = 41.85$, $P < .001$
ETUT (s)	39.57 ± 5.39**	37.44 ± 5.95*	34.68 ± 5.62	$F = 9.77$, $P < .001$
CTUT (s)	48.12 ± 6.96***††	43.13 ± 5.47***	34.87 ± 3.48	$F = 59.35$, $P < .001$

Abbreviations: CS2, cluster sets of 2; CS4, cluster sets of 4; TS, traditional; TUT, total time under tension; ETUT, eccentric time under tension; CTUT, concentric time under tension.

Holm sequential Bonferroni follow-up tests revealed different from TS, $*P \leq .05$, $**P \leq .01$, $***P \leq .001$; different from CS4, $\dagger P \leq .05$, $\dagger\dagger P \leq .01$, $\dagger\dagger\dagger P \leq .001$; and ^no differences.

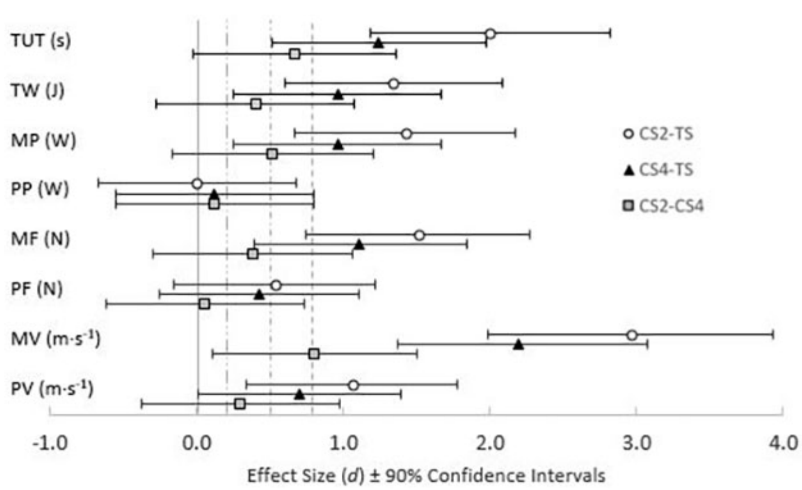

Figure 3 — Effect size ± 90% confidence intervals comparing cluster sets of 2 (CS2), cluster sets of 4 (CS4), and traditional sets (TS). Variables include total time under tension (TUT), total work (TW), mean power (MP), peak power (PP), mean force (MF), peak force (PF), mean velocity (MV), and peak velocity (PV). Dashed lines present at 0.2, 0.5, and 0.8 to indicate small, medium, or large effects of 0.20 to 0.49, 0.50 to 0.79, and >0.80, respectively.

while intending to move at maximal velocity, and (3) using greater loads resulted in less MP but did not affect PP. Therefore, data from the current study show that not only can cluster-set structures be designed to reduce fatigue-induced decreases in velocity compared with traditional-set structures with similar training loads as observed in other studies,[1–4] but they can also be manipulated to increase training load and result in greater TW without negatively affecting PP output.

The application of parallel models of periodization may require a multifaceted approach to training in which the goal is to simultaneously increase strength, hypertrophy, and power output. As increases in muscle mass are associated with increases in strength and strength underpins power output,[14] it can be argued that skeletal-muscle hypertrophy is of great importance. The realization of hypertrophy is a remarkably complex process, but despite the complexity, there is a reoccurring theme in research that focuses on hypertrophy: Protocols that include large amounts of external work (ie, high-volume protocols) generally elicit greater skeletal-muscle hypertrophy than protocols with less external work.[15–17] Ultimately, if a resistance-training program can include greater training loads for a given number of repetitions, increasing TW, it may be plausible that the program would encourage greater skeletal-muscle hypertrophy. Although TW can be increased by increasing the load (ie, force) or the number of repetitions performed (ie, distance), increasing the external load may result in preferential hypertrophy of type II fibers,[18] which can be beneficial for athletic populations that require strength and size. Therefore, the importance of training load for increasing muscle size and strength is apparent.[14,18]

As load increases, the force required to move the barbell increases and a resultant decrease in movement velocity occurs.[7] In line with this force–velocity relationship, high volumes of fatiguing resistance training with heavy loads may be unwarranted for improving external mechanical power output, as movement velocity is reduced, negatively affecting system power output.[19,20] Conversely, cluster sets serve as an alternative method to traditional sets for performing high volumes of resistance training without negatively affecting power output.[1,3,8] The majority of the cluster-set literature includes protocols in which the external load is kept constant between the traditional and cluster-set structures, leaving the influence of cluster sets with various loads unexamined. Therefore, the purpose of this study was to determine the effects of intraset rest frequency and training load on TUT, TW, and mechanical power output during back squats with similar fatigue-induced changes in movement velocity.

The additional intraset rest periods during CS2 allowed subjects to successfully lift a greater external load for all 36 repetitions, resulting in 19% greater MF, 15% greater TW, and 26% greater TUT than TS. In addition, CS4 resulted in 14% greater MF, 10% greater TW, and 16% greater TUT than TS. Despite the importance of TW and training load for muscle growth, other factors also contribute to skeletal-muscle hypertrophy. Previously, researchers have suggested that TUT should be considered when designing resistance-training programs that aim to increase hypertrophy, because increasing the TUT can increase muscle activation.[21,22] Furthermore, it has been suggested that fatigue is necessary for hypertrophy[16,23] and that fatigue is a result of the relationship between TUT and TW.[24,25]

However, as fatigue increases, movement velocity decreases, negatively affecting the power output of the system.[1,3,8,13] Although the absolute velocities were different between protocols in the current study, the %VL from the beginning to the end of each protocol was statistically similar. This fact indicates that all 3 protocols could have been considered equally fatiguing.[13,26] Interestingly, PP was similar between the protocols whereas MP was the least when using the greatest load (CS2) and was the greatest when using the lightest load (TS). Despite the cluster-set protocols' having greater MF (CS2 > CS4 > TS), slower MV (CS2 < CS4 < TS) negatively affected MP (CS2 < CS4 < TS), indicating that MV had a larger effect on MP than MF did. Although PV followed the same pattern as MV (CS2 < CS4 < TS), PP was not different between groups, possibly explained by greater, but not significantly greater, PF experienced in the cluster-set protocols. Nonetheless, the instance at which PP, PF, and PV occurred was not measured in the current study, leaving the aforementioned hypothesis purely speculative.

Our data are in line with those from Oliver et al,[8] who emphasized that mean power output was primarily affected by mean velocity rather than force production. In their study, the total rest time from the traditional-set protocol (4 sets of 10 with 120 s of interset rest) was redistributed to a cluster-set protocol (4 sets of 10 with 90 s of interset rest and 30 s of intraset rest) using ~70% 1RM during parallel back squats. They demonstrated greater mean power outputs with cluster sets than traditional sets. However, in the traditional-set protocol, subjects had to reduce the load by an average of 8% to complete all 40 repetitions, and they displayed greater decreases in mean velocity than with the cluster-set protocol, suggesting that the cluster sets were less fatiguing. That study[8] supports the notion that modifying a set structure to include more-frequent rest periods may result in better maintenance in velocity than traditional sets, which is beneficial in the practical realm of strength and conditioning. However, when unplanned changes to the external load occur during a scientific protocol, the load becomes an uncontrolled, independent variable, which may be perceived as undesirable.

To address the concerns of unintentional changes in external load, other researchers[3] used the same protocols as the ones in the current study with the exception of all protocols using 60% 1RM, and unlike the subjects in the study conducted by Oliver et al,[8] did not have to reduce the load during any of the protocols. As a result, mean force was not different between any of the protocols, but mean velocity and power output were least during traditional sets and greatest during cluster sets with the most intraset rest.[3] The results of Tufano et al[3] agree with the findings of Oliver et al,[8] showing that when mean force is similar, movement velocity is the main determinant of mean power output. However, data from the current study show that even when MF differs in response to intentional alterations in external load, MV still appears to have the greatest influence on MP.

Finally, a look at the available cluster-set literature shows that many authors discuss either peak[5,6,27] or mean power system output[1,8,28] but rarely report both.[3] It is debatable whether MP or PP is more important for different athletes, but the ability to apply large forces at high velocities is widely accepted to be important in many sports. The differing responses between MP and PP in the current study may inspire other researchers to investigate the relationship of PP and MP to various sport performance tasks and resistance-training adaptations.

While CS2 and CS4 appear to be preferable for increasing acute mechanical hypertrophic factors, physiological factors that influence muscle growth were not measured in the current study. It is known that metabolic stress, in addition to mechanical stress, also contributes to skeletal-muscle growth.[15,22] While not within the scope of the current study, it is possible that an increase in total rest time during CS2 and CS4 may have affected metabolite accumulation and energy availability by sparing phosphagen energy stores and decreasing the reliance on glycolytic energy pathways.[29,30] Therefore, is it possible that although CS2 and CS4 increased mechanical stress compared with TS, additional recovery during the intraset rest periods may have limited metabolic stress, possibly negating the benefits of increased TW and TUT. Since %MVL and %PVL were not different between protocols in the current study, it could be hypothesized that ATP-PCr energy substrates were expended and restored at similar rates between the protocols based on previous research linking fatigue-induced decreases in velocity to the energetic demands of an exercise.[13,29,30] If metabolic stress was in fact similar between protocols but CS2 and CS4 resulted in greater TUT and TW, these cluster-set protocols may provide a superior stimulus for developing skeletal-muscle hypertrophy than TS without sacrificing PP. However, we did not measure these variables in the current study, warranting the investigation of the mechanisms underlying the force-velocity-power fatigue phenomenon.

Another limitation of the current study is the difference in session duration. By design, the protocol duration for CS2 was greater than for CS4, both of which took longer than TS (15:51 ± 0:49, 10:10 ± 0:38, and 6:01 ± 0:19 min, respectively). Therefore, if CS2 is applied across multiple exercises within a training session, either the session's duration would be much longer than TS or the number of exercises performed in the same amount of time would be fewer during CS2. Although CS2 and CS4 resulted in greater TW and TUT than TS in the current study, it could be argued that CS2 and CS4 were less efficient than TS since TW was 15% and 10% greater than TS, but session duration was about 164% and 69% longer during CS2 and CS4, respectively. To abridge these values, the mean training efficiency (total work per unit of time) during TS was 12.8 kJ/min whereas CS2 and CS4 resulted in 5.5 and 8.3 kJ/min, respectively. Therefore, subjects may have been able to perform 1 or 2 additional sets during TS, increasing mechanical stress. However, this was not investigated in the current study because increasing the number of sets and repetitions in any of the protocols might have affected %VL, resulting in non-fatigue-matched protocols. Nonetheless, the application of such protocols in training environments allows for ad hoc flexibility that is not allowed in a scientific setting.

Although %VL was similar between protocols, the different loads used in each set structure resulted in different absolute velocities, which may alter the transfer to performance. Keeping this in mind, it is the responsibility of the strength and conditioning professional to determine the importance of absolute velocity during a resistance-training session and to understand that the cluster-set structures in the current study were specifically designed to determine whether cluster-set structures could be manipulated to increase TW and TUT while maintaining PP output, with no concern for absolute movement velocity.

To conclude, our data show that cluster sets can be manipulated in various ways that encourage different acute training responses, not just to result in less fatigue as demonstrated by previous studies.[1,3,8,28] Strength and conditioning professionals must be cognizant of the effects of external loads and rest-period frequencies on velocity, fatigue, and system power output to prescribe training recommendations geared toward various adaptations, and further research should investigate the effects of different loads and intraset rest durations and frequencies in cluster-set structures.

Practical Applications

The current study shows that cluster sets can be manipulated to include greater training loads, resulting in greater TW and similar PP but less MP without increasing %MVL or %PVL compared with traditional sets. When designed in this manner, cluster sets may serve as an alternative to traditional program design for promoting muscle growth over time during parallel periodization models. The duration of the CS2 session was longer than CS4 and TS due to more frequent intraset rest intervals, but this allowed for CS2 to have the greatest training load, TW, and TUT. Therefore, the CS4 structure may be warranted when aiming to increase TW and TUT under brief, but not strict, time constraints. Finally, if the maintenance of MP is of greater importance than PP, caution should be used when utilizing heavier loads during high-volume training, even if cluster sets are used.

Conclusions

The current study shows that if minimizing fatigue-induced changes in velocity is not of paramount importance, cluster-set structures can be designed to use greater loads for a given number of repetitions, resulting in greater TW without sacrificing PP. However, it is important that the strength and conditioning professional decide whether maximizing velocity or maximizing mechanical stress is most important during training. Future research should investigate the effects of similar set structures on other acute hypertrophic variables such as muscle activation, metabolic responses, and endocrine responses, ultimately investigating such cluster-set structures over a chronic training period.

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